

Chaos Lab Build Guide & Operator's Manual

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Scope: Quantum Chaos Enterprise Lab (Chaos Lab) — local simulated enterprise for crypto discovery and reporting validation.

1. Purpose

Chaos Lab is a simulated enterprise environment for validating crypto discovery (TLS posture, certificate hygiene, mTLS behavior), service inventory enrichment (protocol classification, non-TLS services), and enterprise-ish topology (ingress + SNI, identity + PKI, cloud simulators). It is built to generate meaningful scanner findings and reporting datapoints for quantum readiness assessments.

2. What's Included

- Core: modern/legacy/expired/self-signed TLS, mTLS-required, HTTP (8000), HTTP-on-8444, SSH (2222), unknown services (5555/5556), slow handshake proxy (12443).
- Phase A (phaseA): service inventory expansion, TLS chain scenarios (missing intermediate, RSA 1024, SHA1), ingress/SNI multi-vhost behind 24443.
- Phase B (cloud): LocalStack (S3+STS+IAM) and Azurite (blob/queue/table) behind TLS fronts with SNI hostnames.
- Phase C (identity+pkc): Keycloak + Postgres, step-ca, mTLS gateway mtlc-stepca.chaos.local:17443, cert rotation automation.

3. Prerequisites

- Docker Desktop (macOS or Windows) + Docker Compose v2
- openssl on host, curl recommended

4. Quick Start

Validate compose, start core, or start everything with profiles.

```
docker compose -p chaoslab -f docker-compose.yml config >/dev/null && echo "compose OK"
```

```
docker compose -p chaoslab -f docker-compose.yml up -d
```

```
docker compose -p chaoslab -f docker-compose.yml \
--profile phaseA --profile cloud --profile identity --profile pki up -d
```

```
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
```

5. Operator's Manual

Common operations for logs, restarts, stop/reset.

```
docker logs <container_name> --tail 100
```

```
docker compose -p chaoslab -f docker-compose.yml restart <service>
```

```
docker compose -p chaoslab -f docker-compose.yml down
```

```
docker compose -p chaoslab -f docker-compose.yml down -v # deletes volumes
```

6. Phase C (PKI / mTLS)

Generate step-ca artifacts and restart the Phase C gateway so nginx serves newly issued certificates. Loop mode re-issues on an interval.

```
chmod +x scripts/phaseC_stepca_issue.sh
./scripts/phaseC_stepca_issue.sh
# loop:
./scripts/phaseC_stepca_issue.sh --loop --every-min 5
```

Validate mTLS fail/succeed:

```
curl --cacert certs/stepca/ca-bundle.crt \
--resolve mtls-stepca.chaos.local:17443:127.0.0.1 \
https://mtls-stepca.chaos.local:17443/__tls_ok -v

curl --cacert certs/stepca/ca-bundle.crt \
--cert certs/stepca/client1-chain.crt \
--key certs/stepca/client1.key \
--resolve mtls-stepca.chaos.local:17443:127.0.0.1 \
https://mtls-stepca.chaos.local:17443/__tls_ok
```

Verify served server cert (serial changes after rotation):

```
openssl s_client -connect 127.0.0.1:17443 -servername mtls-stepca.chaos.local
</dev/null 2>/dev/null \
| openssl x509 -noout -dates -serial
```

7. Cloud (Phase B)

LocalStack health via TLS front and Azurite TLS fronts.

```
curl -k --resolve aws.chaos.local:24566:127.0.0.1
https://aws.chaos.local:24566/__tls_ok
curl -k --resolve aws.chaos.local:24566:127.0.0.1
https://aws.chaos.local:24566/_localstack/health

curl -k --resolve blob.chaos.local:21000:127.0.0.1
https://blob.chaos.local:21000/__tls_ok
curl -k --resolve queue.chaos.local:21001:127.0.0.1
https://queue.chaos.local:21001/__tls_ok
curl -k --resolve table.chaos.local:21002:127.0.0.1
https://table.chaos.local:21002/__tls_ok
```

8. Troubleshooting

- Port conflicts: stop local services, change host mappings, or use -p chaoslab2.
- Profiles: Phase C typically needs both --profile identity and --profile pki.
- Rotation but no change: recreate mtls-stepca-gateway to load new cert files.
- Issuer errors: use --cacert certs/stepca/ca-bundle.crt and ensure fullchain is served.

9. Safety Notes

Chaos Lab is intended for local testing. Avoid exposing ports publicly. Treat private keys as sensitive even in lab contexts.

10. Next Roadmap

- Enterprise realism: client2/client3, allow/deny rules, break-glass cert role.
- Reporting: richer service inventory outputs, mTLS detection, issuer/chain visibility, cert TTL scoring.